Computer Physics Communications 150 (2003) 235–246

Computer Physics Communications

www.elsevier.com/locate/cpc

Calculating modes of quantum wire and dot systems using a finite differencing technique

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Received 7 May 2002; received in revised form 24 July 2002

Abstract

In this paper the Schrödinger equation of both a quantum wire and a quantum dot are solved using a finite difference approach. It is demonstrated that the method is valid for the simple case of an infinitely deep quantum wire, where the solutions obtained are within 0.25 meV of the analytical solutions. The method is then used to calculate the eigenenergies of a triangular wire with finite barriers. The eigenenergies of the more complex case of a pyramidal quantum dot were then calculated using this method. The method is compared to an eigenvalue method in terms of memory usage, time requirements and the numerical solutions. It is shown that this method has the advantages of being relatively fast, usable with any wire geometry and any potential profile. In addition, the demand on computer memory varies linearly with the size of the system under investigation.

Keywords: Quantum wire; Finite difference; Sparse matrices; Pyramidal quantum dot; Energy levels

1. Introduction

With the advent of recent advances in epitaxial crystal growth technology, such as molecular-beam epitaxy [1], which have enabled the fabrication of atomically sharp heterojunction interfaces, there has been a growing interest in spatially quantized systems [2]. Of these systems, quantum well wires (QWW) have attracted much attention. It has been shown that transistors and lasers fabricated from quantum wires exhibit excellent characteristics [3], while quantum dots (QD) have recently been utilized in the construction of photomixer devices used for terahertz generation [4], in photoconductive intersubband detectors [5] and even in lasers [6].

As with all quantized systems the cornerstone of any theoretical analysis or practical implementation of a QWW or QD is the solution of Schrödinger's equation. A variety of particular cases of this equation have been solved using a wide range of numerical methods. These include: expressing the wave function as a series of orthogonal periodic functions [7,8], Fourier expansion [9,10], plane wave expansion [11], **k.p** theory [12] and finite differencing [13]. The methods based on Fourier-type expansions suffer when moving to higher dimensions. In one dimension a basis set of N elements, becomes N^2 and N^3 in two and three dimensions, respectively. Direct diagonalisation in three

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PII: S0010-4655(02)00690-2

dimensions therefore has a matrix with N^6 elements. This is computationally intensive in terms of both memory and CPU processing time.

In this paper a finite difference method is developed for the solution of the time-independent, constant effective mass Schrödinger equation. The motivation behind this approach is to develop a computational technique which is fast to execute and light on memory. The ultimate goal is to consider very large and complex three-dimensional systems, such as arrays of embedded quantum dots in doped semiconductor wells. The technique therefore has to be readily expandable to larger spatial areas and general three-dimensional potentials. The technique presented here does indeed have these properties.

For simplicity and dissemination, the method is developed for QWWs of any cross-section. An overview of the theory of this method is presented in this paper, along with results and comparison with existing data [7] in the case of a triangular potential profile. The method is also used to solve the more complex case of a pyramidal QD, and a comparison made to an eigenvalue method [14] in terms of memory usage, time requirements as well as the numerical solutions.

2. Theory: finite difference expansion

The system under investigation has the simplest possible geometry. It consists of an infinitely deep rectangular cross-sectional wire. While this geometry is not really practical, it nevertheless provides a good starting point since this method should be versatile enough to solve for any cross-section and any potential. The quantum wire used is shown schematically in Fig. 1.

In this initial work, attention is focused on the general three-dimensional constant effective mass Schrödinger equation given by

$$\frac{-\hbar^2}{2m^*}\nabla^2\psi(x,y,z) + V(x,y,z)\psi(x,y,z) = E\psi(x,y,z), \tag{1}$$

where m^* the electron effective mass, and $V(x,y,z)$ the potential energy. If the potential is only a function of y and z then the motion along the length x of the wire can be decoupled [15], thus leading to a one-dimensional Schrödinger equation for the motion along the x direction which can be solved analytically, and is given by

$$\frac{-\hbar^2}{2m^*}\frac{\partial^2\psi(x)}{\partial x^2} = E_x\psi(x) \tag{2}$$

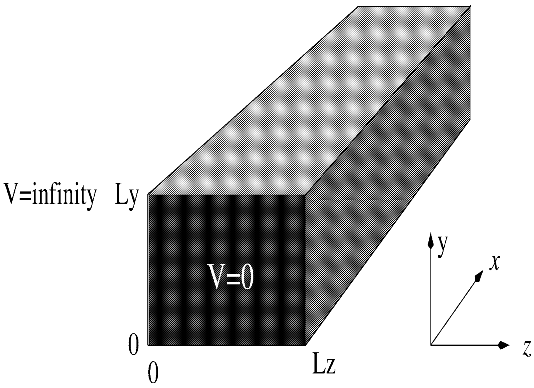

Fig. 1. Infinitely deep rectangular cross-section quantum wire.

and a two-dimensional Schrödinger equation for the confinement within the cross-section of the wire which is our main concern, given by

$$\frac{-\hbar^2}{2m^*}\left[\frac{\partial^2\psi(y,z)}{\partial y^2}+\frac{\partial^2\psi(y,z)}{\partial z^2}\right]=\big(E_{y,z}-V(y,z)\big)\psi(y,z), \tag{3}$$

where the full solution to Eq. (1) is given by

$$\psi(x,y,z)=\psi(x)\psi(y,z). \tag{4}$$

Expanding Eq. (3) in terms of finite differences, while ensuring that the step lengths δy and δz are sufficiently small so that the approximation is valid, yields

$$\begin{aligned}&(\delta y)^2\big[\psi(y,z+\delta z)+\psi(y,z-\delta z)\big]+(\delta z)^2\big[\psi(y+\delta y,z)+\psi(y-\delta y,z)\big]\\&\quad-\left[\frac{-2m^*}{\hbar^2}\big(E_{y,z}-V(y,z)\big)(\delta y\delta z)^2+2\big((\delta y)^2+(\delta z)^2\big)\right]\psi(y,z)=0.\end{aligned} \tag{5}$$

In order to solve Eq. (5) numerically for $\psi(y,z)$, it is necessary to discretize the cross-section of the wire into blocks of length δz and height δy. The wave function is now mapped to the elements of this two-dimensional array. These array elements can now be labeled in the usual way. The value of δz is chosen so that the total width of the wire, L_z, is represented by the total number of columns, n_1+2. Similarly, δy is chosen so that the total height of the wire, L_y, is represented by the total number of rows, n_2+2. This arrangement is shown schematically in Fig. 2. For ease of notation, the value of the wave function at each array point is now labeled by its corresponding row and column indices, i.e. (i,j). Rewriting Eq. (5) in terms of these indices yields

$$(\delta y)^2(\psi_{i,j+1}+\psi_{i,j-1})+(\delta z)^2(\psi_{i-1,j}+\psi_{i+1,j})-k\psi_{i,j}=0, \tag{6}$$

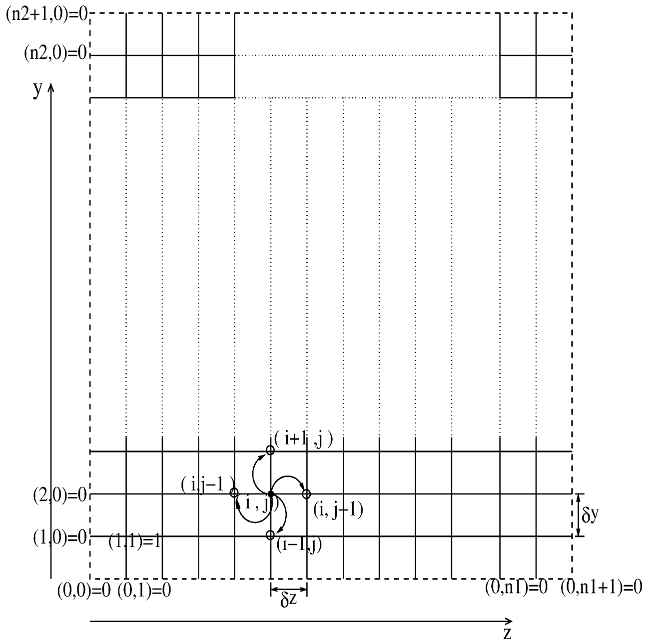

Fig. 2. Schematic illustration of array representing wire and iterations.

where

$$k = \frac{-2m^*}{\hbar^2}\big(E_{y,z} - V(y,z)\big)(\delta y \delta z)^2 + 2\big((\delta y)^2 + (\delta z)^2\big). \tag{7}$$

Eq. (6) shows that the value of the wave function at each point depends on the four surrounding grid points. Hence, the wave function can be calculated for any energy by solving the sets of simultaneous equations generated by Eq. (6), provided that an appropriate perturbation value is chosen. This perturbation has to be a non-zero value, otherwise the solution to this system of linear equations would be uniformly zero. Since scaling the wave function by a constant factor will not affect the energy eigenvalues, it is then reasonable to choose the following perturbation value

$$\psi_{1,1} = 1. \tag{8}$$

2.1. Matrix equation

The method developed depends on the ability to write the simultaneous equations as a single matrix equation. The matrix equation has the form of

$$\mathbf{A}.\mathbf{\Psi} = \mathbf{S}, \tag{9}$$

where **A** is the matrix of wave function coefficients and $\mathbf{\Psi}$ is a column vector of wave function points. **S** is a column vector consisting of the source terms. They are called the source terms in analogy with a heat transfer equation [16] which is solved in a similar manner. In the present case the source is replaced by the perturbation value. The standard boundary conditions, given below, are satisfied by all solutions.

$$\psi(y,z) \to 0 \quad \text{as } z \to \pm\infty, \tag{10}$$

$$\psi(y,z) \to 0 \quad \text{as } y \to \pm\infty. \tag{11}$$

This is accomplished by forcing all the outermost array elements to zero (shown as the dashed lines in Fig. 2). The computational domain was chosen to be large enough to ensure that the derivatives and the consequent eigenvalues were independent of the computational domain. Thus, it is only necessary to calculate the internal points of the array thereby reducing the number of unknowns to $[n_1 \times n_2]$. Also, since $\psi_{1,1}$ is defined by the perturbation value of Eq. (8), there remains only $[n_1 \times n_2] - 1$ points to calculate. In order to further simplify the problem, $\delta z = \delta y$ is assumed.

Thus the column vector $\mathbf{\Psi}$ will consist of $n = [n_1 \times n_2] - 1$ elements, which is the total number of unknown elements and is given by Eq. (12). It consists of a total of n_2 blocks each of length n_1 elements, with the exception of the first block which consists of $n_1 - 1$ elements since $\psi_{1,1}$ is already determined

$$\mathbf{\Psi} = \begin{pmatrix} \psi_{1,2} \\ \psi_{1,3} \\ \vdots \\ \psi_{1,n_1} \\ \psi_{2,1} \\ \vdots \\ \psi_{n_2,n_1} \end{pmatrix}. \tag{12}$$

The matrix $\mathbf{A}$ will be a $(n \times n)$ sparse matrix. The best way to represent this matrix, is using sub-matrices. Here, there are n_2 blocks or sub-matrices, which can be written as follows:

$$\mathbf{A} = \overbrace{\begin{pmatrix} \beta_1 & I_1 & \mathbf{0} & \cdots & \cdots & \mathbf{0} \\ I_2 & \beta & I & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & I & \beta & I & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \mathbf{0} \\ \mathbf{0} & \cdots & \mathbf{0} & I & \beta & I \\ \mathbf{0} & \cdots & \cdots & \mathbf{0} & I & \beta \end{pmatrix}}^{n_2 \text{ blocks}}, \tag{13}$$

where the sub-matrix β_1 is a $([n_1 - 1] \times [n_1 - 1])$ tridiagonal matrix of the form:

$$\beta_1 = \begin{pmatrix} -k & 1 & 0 & \cdots & 0 \\ 1 & -k & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & -k & 1 \\ 0 & \cdots & 0 & 1 & -k \end{pmatrix}. \tag{14}$$

The remaining β sub-matrices all have the form of Eq. (14), but are all $(n_1 \times n_1)$ matrices. The sub-matrix I_1 is a $([n_1 - 1] \times n_1)$ matrix of the form:

$$I_1 = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \end{pmatrix}. \tag{15}$$

The sub-matrix I_2 is a $(n_1 \times [n_1 - 1])$ matrix of the form:

$$I_2 = \begin{pmatrix} 0 & \cdots & \cdots & 0 \\ 1 & 0 & \cdots & \vdots \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{pmatrix}. \tag{16}$$

All the sub-matrices I are $(n_1 \times n_1)$ identity matrices. The remainder of the matrix $\mathbf{A}$ is comprised of zeroes.

Finally, the matrix $\mathbf{S}$ is a column vector of n elements and is of the form:

$$\mathbf{S} = \begin{pmatrix} -\psi_{1,1} \\ 0 \\ \vdots \\ 0 \\ -\psi_{1,1} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \begin{matrix} \\ \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} \begin{matrix} n_1-2 \\ \text{zeroes} \end{matrix} \\ \\ \\ \\ \\ \end{matrix}. \tag{17}$$

Having established the general forms of the matrices involved, there are several computer packages available which can be used to solve this system of equations. In this case the package MATLAB® was used.

2.2. Automation

The only issue that remains to be addressed is the identification of the correct physical solutions; since in spite of the fact that a mathematical solution can be calculated for all energies, correct physical solutions only occur at certain energies. In the one-dimensional shooting method [15] the correct physical solutions are identified by finding the solutions that occur for energies E, which satisfy the standard boundary conditions of Eqs. (10) and (11). In the case of this matrix method it is not possible to use the boundary conditions to identify the eigenenergies, see Section 2.1.

The new aspect introduced in this work is to base the search for the correct physical solutions on the energies where the wave function has well defined maxima away from the edges of the y–z plane. The exact locations of these maxima depend on the character of the eigenstate, e.g., for the ground state the maxima should be at the centre of the plane. The initial condition, stated in Eq. (8), of setting one of the corner values to unity, ensured that away from a physical solution, the maxima across the plane was always at this point. Only when the scanned energy was close to a solution did a local maxima occur away from this point. The correct energies were identified by plotting the position of the maxima in the y–z plane versus the energy. The resulting plot not only gives a good estimate for the value of the eigenenergy but also gives a good indication of the character of the eigenstate, i.e. ground state, first excited state, etc. Fig. 3 shows a plot of position of the maxima along the y-axis of the QWW versus energy for an infinitely deep 600×600 Å QWW. The well defined delta-like spikes indicate the energy level of the different states. Figs. 4 and 5 show the wave functions at the energies of the first two spikes. Table 1 shows

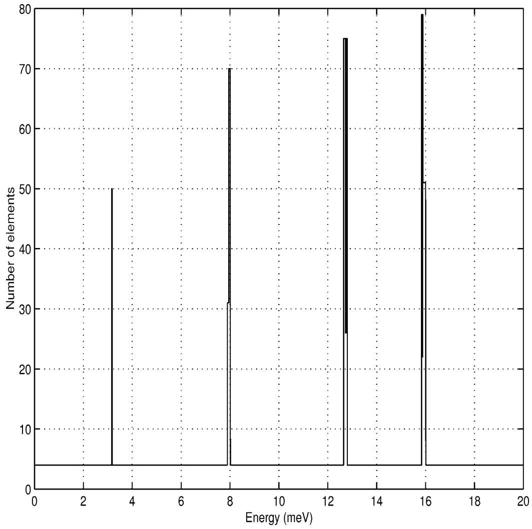

Fig. 3. Position of maxima along the y-axis vs. energy for a 600×600 Å wire with zero potential.

Table 1
Comparison of results between analytical and numerical values for a 600×600 Å infinite QWW

State	Numerical values (meV)	Analytical values (meV)
Ground	3.18	3.12
First	7.95	7.95
Second	12.65	12.47
Third	15.83	15.59

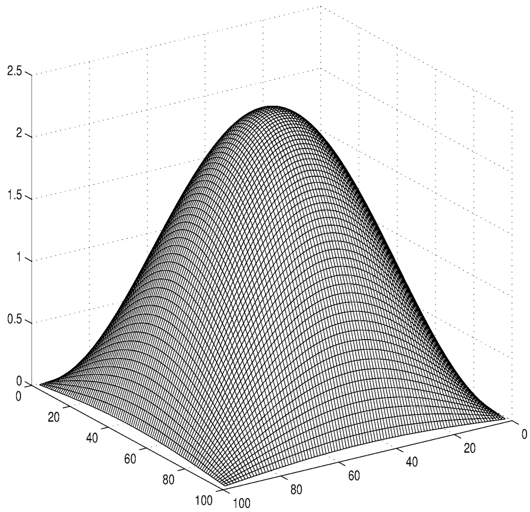

Fig. 4. Unnormalized ground state wave function for a 600 × 600 Å wire with zero potential, horizontal coordinates represent position on the cross-section of the QWW. Energy = 3.18 meV.

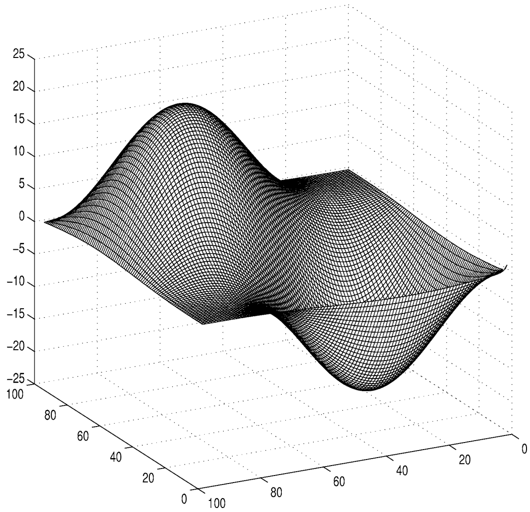

Fig. 5. Unnormalized wave function of the first excited state for a 600 × 600 Å wire with zero potential, horizontal coordinates represent position on the cross-section of the QWW. Energy = 7.951 meV.

the eigenenergies, obtained using this method, of the first four states compared to the eigenenergies, of the same eigenstates, which were calculated analytically [15]. The mesh used for these calculations was a 100 × 100 mesh, i.e. a mesh separation of 6 Å. As can be seen, the calculated eigenenergies lie within 0.25 meV of the analytical values.

2.3. Extension to three dimensions

Having deduced the form of the equations and matrices needed for the two-dimensional solution, it is a relatively straightforward matter to extend the method to three dimensions.

Consider the simple case of a quantum box of dimensions L_x, L_y and L_z. The box can discretized in such a way that, by choosing appropriate scales, the dimensions of the box can be represented by a total of $n_1 + 2$ elements for the y-dimension, $n_2 + 2$ for the z-dimension and $n_3 + 2$ for the x-dimension. Choosing $\delta x = \delta z = \delta y$ it can be shown, in a similar manner to that of the two-dimensional case, that the resulting equations can be simplified to

$$\psi_{i,j,l+1} + \psi_{i,j,l-1} + \psi_{i,j-1,l} + \psi_{i,j+1,l} + \psi_{i+1,j,l} + \psi_{i-1,j,l} - k\psi_{i,j,l} = 0 \tag{18}$$

and

$$k = \frac{-2m^*}{\hbar^2}\big(E_{x,y,z} - V(x,y,z)\big)(\delta y)^2 + 6, \tag{19}$$

where i, j, l are the array indices of the wave function points. The perturbation value selected was

$$\psi_{1,1,1} = 1. \tag{20}$$

As before, this set of equations can be converted to a two-dimensional array, **A**, of wave function coefficients, a column vector, $\boldsymbol{\Psi}$, of wave function points and a column vector, **S**, of source terms. Again, all solutions satisfy the standard boundary conditions

$$\psi(x,y,z) \to 0 \quad \text{as } x \to \pm\infty, \tag{21}$$

$$\psi(x,y,z) \to 0 \quad \text{as } y \to \pm\infty, \tag{22}$$

$$\psi(x,y,z) \to 0 \quad \text{as } z \to \pm\infty. \tag{23}$$

The column vector $\boldsymbol{\Psi}$, is of length n, where $n = [n_1 \times n_2 \times n_3] - 1$. It consists of a total of n_1 blocks each of length $n_2 \times n_3$ elements.

The matrix **A** will be an $n \times n$ sparse matrix. It has the same form as that of the two-dimensional case, except that it has n_1 blocks of length $n_2 \times n_3$ elements.

The vector **S** is of length n and is of the form

$$\mathbf{S} = \begin{pmatrix} -\psi_{1,1,1} \\ 0 \\ \vdots \\ 0 \\ -\psi_{1,1,1} \\ 0 \\ \vdots \\ 0 \\ -\psi_{1,1,1} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \begin{matrix} \\ \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} {\scriptstyle n_3-1 \text{ zeroes}} \\ \\ \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} {\scriptstyle n_2\times n_3 \text{ zeroes}} \\ \\ \\ \\ \\ \end{matrix} . \tag{24}$$

The automation process for this case is identical to that of two dimensions.

3. Results and discussion

Having demonstrated the validity of the method for the simple case of an infinitely deep wire, it will now be used to calculate the eigenenergies of more complicated structures.

The first system to be examined was a finite barrier triangular quantum wire which was taken from the literature [7] for ease of comparison. The QWW used consisted of a triangular zero potential region embedded

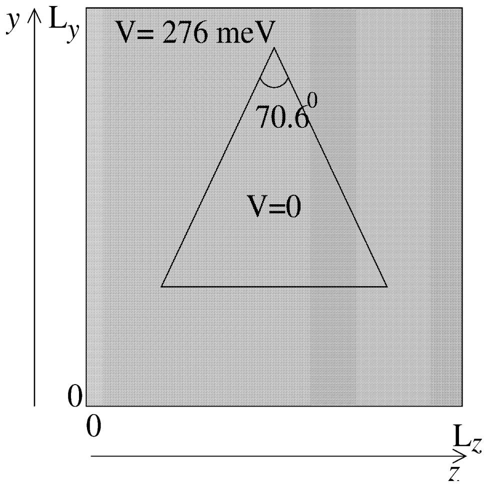

Fig. 6. Figure showing rectangular cross-sectional wire with finite barrier height with triangular region having no potential.

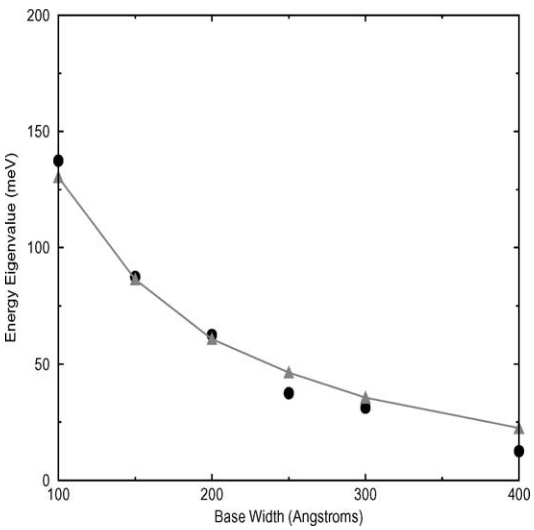

Fig. 7. Energy eigenenergies of triangular wire with finite barrier potential. Dots data from Gangopahdhyay [7], line with triangle data obtained using the finite difference method presented here.

in a 600 × 600 Å rectangular cross sectional wire. The triangle was an isosceles triangle with an angle of 70.6° subtended by the two equal sides. The values of the parameters were chosen to correspond with those in the literature [7]. The values used were: $m^* = 0.067m_0$, and $V = 276$ meV, see Fig. 6.

Different values for the base width and height of the triangle were used in the calculation of the energy states. Fig. 7 shows the results obtained using this method along with the data found in the literature.

The results obtained from this method show very good agreement with that of Gangopahdhyay [7], with the difference in eigenenergies ranging between two and ten meV.[1]

[1] This is likely to arise from the simplified constant effective mass Hamiltonian employed here for this initial demonstration.

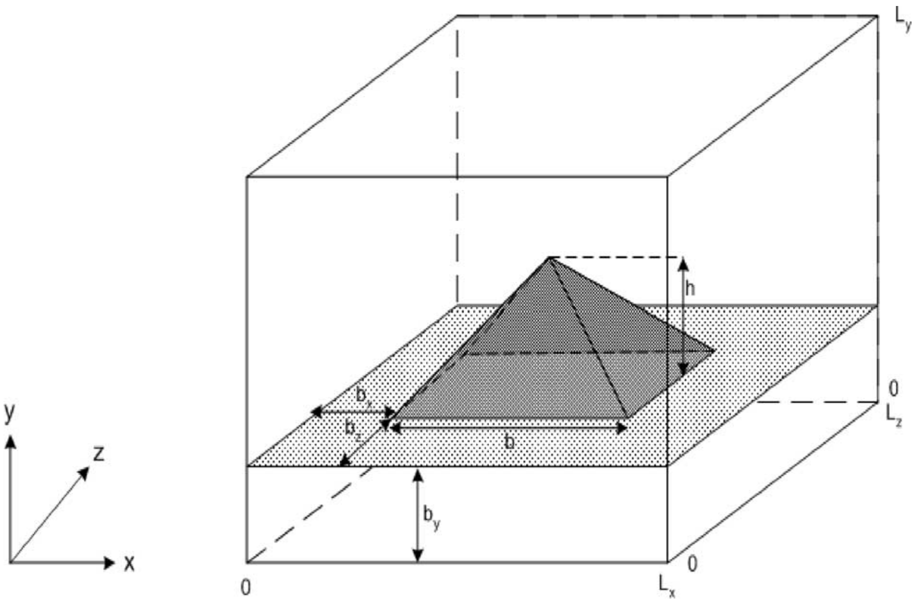

Fig. 8. Schematic of pyramidal quantum dot.

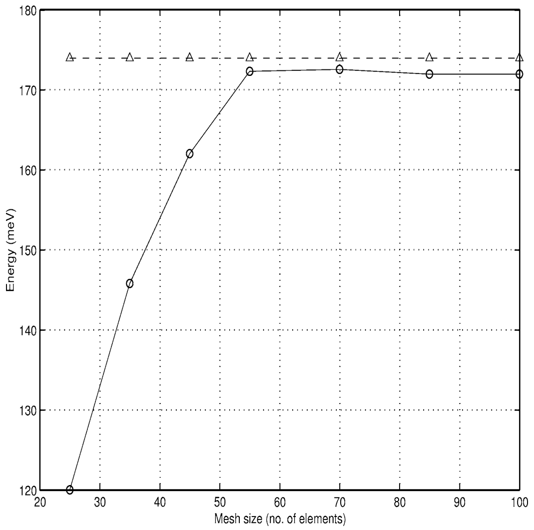

Fig. 9. Energy vs. Mesh size for pyramidal QD. Line with triangles shows data from literature [14], line with circles data obtained using finite difference method.

The second system that was investigated was a pyramidal quantum dot, the dimensions and parameters of which were chosen to closely correspond with those used by Califano [14] for ease of comparison. A schematic of this system is displayed in Fig. 8. The values of the parameters were: $m^* = 0.0665m_0$, $V = 276$ meV, $L_x = L_y = L_z = 520$ Å, $b = 120$ Å, $h = 60$ Å and $b_x = b_y = b_z = 200$ Å.

The eigenenergies were calculated using a number of different mesh sizes to ensure convergence. Fig. 9 shows the ground state energy vs. mesh size, along with the numerical value found in the literature [14]. It can clearly be seen that the energy converges as the mesh size is increased.

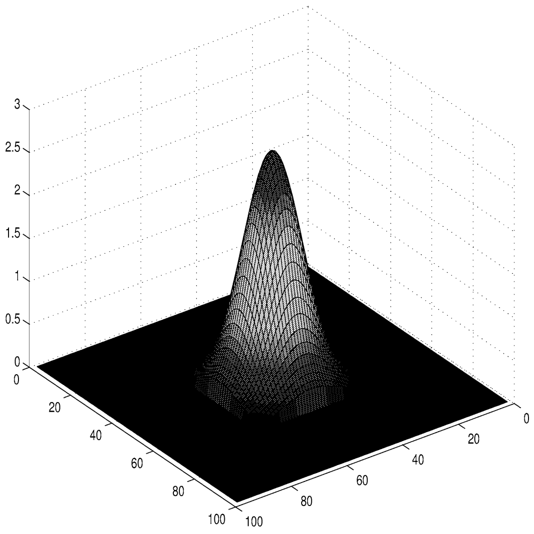

Fig. 10. Slice, at $z = 5.2$ Å, of unnormalized ground state wave function for a $520 \times 520 \times 520$ Å pyramidal quantum dot with potential barriers of 276 meV, horizontal coordinates represent position on the x–y plane of the QD. Energy = 172.15 meV.

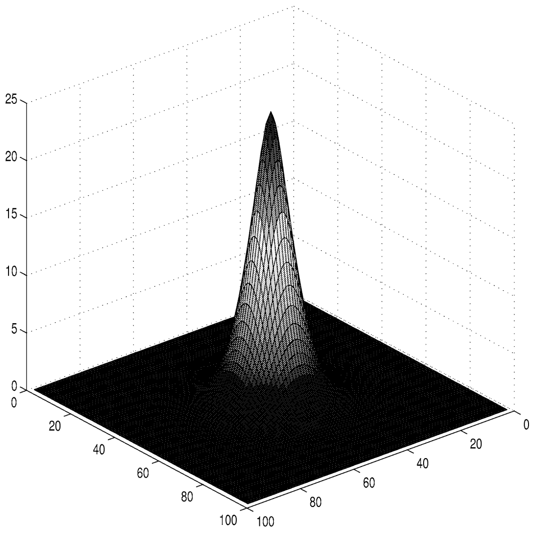

Fig. 11. Slice, at $z = 260$ Å, of unnormalize ground state wave function for a $520 \times 520 \times 520$ Å pyramidal quantum dot with potential barriers of 276 meV, horizontal coordinates represent position on the x–y plane of the QD. Energy = 172.15 meV.

Since the wave functions obtained from these calculations are three-dimensional it was deemed necessary to plot slices of the wave function, i.e. plotting the x–y plane at a constant value of z. Figs. 10 and 11 show two such slices, where the mesh size used for the calculations was $100 \times 100 \times 100$ elements, i.e. a mesh spacing of 5.2 Å.

The eigenvalue method with which a comparison was made, requires 1 Gigabyte of memory just to store the matrix [14]. The finite difference method requires only 7 Megabytes to store the matrix, due to the fact that the matrix is a sparse one. In terms of computation time, the eigenvalue method required, at best, just under 7 hours to reach a solution. The finite difference method requires a significantly smaller amount of computational time, since a single iteration (over energy) requires about 38 minutes (on a 400 MHz PC) and a solution can be found within a few iterations, depending on the initial guess.

All results shown here assume a constant effective mass. To include the differences in effective mass across boundaries return to the original Schrödinger equation given by (1). Note that the kinetic energy operator contains the term $\frac{1}{m^*}\nabla^2$, this now becomes $\nabla\frac{1}{m^*}\nabla$ [15]. The generation of the finite difference expansions is then straightforward. The new matrix equations are more complicated but follow in the same manner as those shown in this paper.

4. Conclusion

A numerical finite difference method was developed and used to calculate the eigenenergies of an infinitely deep quantum wire, and a triangular quantum wire. In the case of the infinitely deep quantum wire the numerical values agree with the analytical values to within 0.25 meV. In the case of the triangular wire the agreement was to within 10 meV to the values found in the literature [7]. For the case of the pyramidal quantum dot the agreement was within 2 meV of the numerical values found in the literature [14].

This method has the advantages of being relatively light on memory, requiring a small fraction of the memory requirement of an eigenvalue method [14], and being relatively fast, where a solution can be found in approximately half the time needed by the eigenvalue method. In addition, this finite difference method is usable with any wire or dot geometry and any potential profile.

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